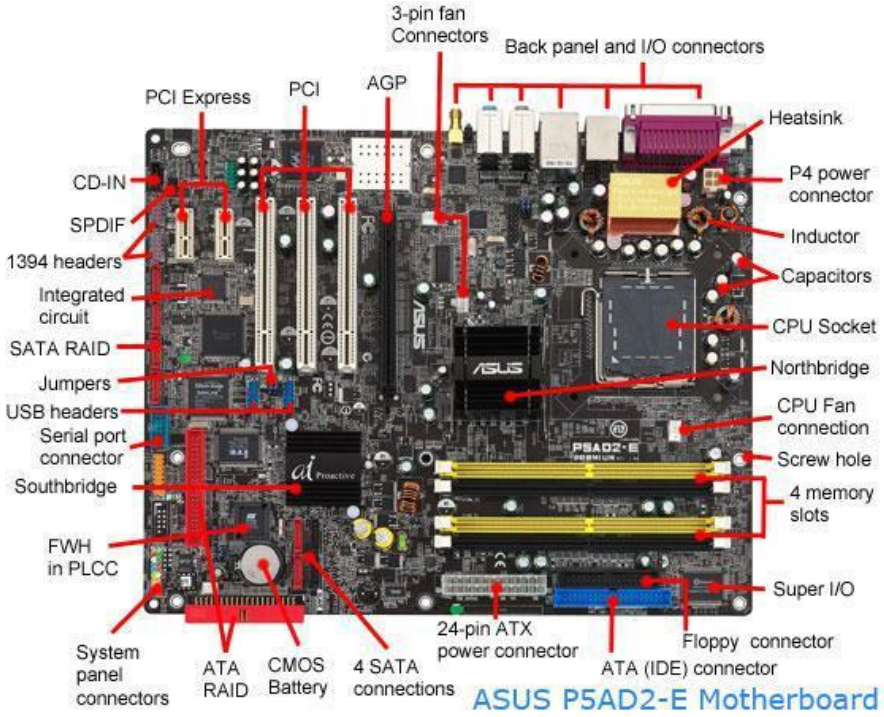
MAJOR COMPONENTS OF COMPUTER SYSTEM

Mother board

A motherboard provides connectivity between the hardware components of a computer, like the processor (CPU), memory (RAM), hard drive, and video card. There are multiple types of motherboards, designed to fit different types and sizes of computers.Each type of motherboard is designed to work with specific types of processors and memory, so they don't work with every processor and type of memory. However, hard drives are mostly universal and work with the majority of motherboards, regardless of the type or brand.



RAM modules

RAM(Random Access Memory) is a part of computer's Main Memory which is directly accessible by CPU. RAM is used to Read and Write data into it which is accessed by CPU randomly. RAM is volatile in nature, it means if the power goes off, the stored information is lost. RAM is used to store

the data that is currently processed by the CPU. Most of the programs and data that are modifiable are stored in RAM.
Integrated RAM chips are available in two form:
1. *SRAM(Static RAM)* : Static memories(SRAM) are memories that consist of circuits capable of retaining their state as long as power is on. Thus this type of memories is called volatile memories.

2. *DRAM(Dynamic RAM)*: DRAM stores the binary information in the form of electric charges that applied to capacitors. The stored information on the capacitors tend to lose over a period of time and thus the capacitors must be periodically recharged to retain their usage. The main memory is generally made up of DRAM chips.

Types of DRAM:
There are mainly 5 types of DRAM:
1. *Asynchronous DRAM (ADRAM)*: The DRAM described above is the asynchronous type DRAM. The timing of the memory device is controlled asynchronously. A specialized memory controller circuit generates the necessary control signals to control the timing.

2. *Synchronous DRAM (SDRAM)*: These RAM chips' access speed is directly synchronized with the CPU's clock. For this, the memory chips remain ready for operation when the CPU expects them to be ready.

3. *Double-Data-Rate SDRAM (DDR SDRAM)*: This faster version of SDRAM performs its operations on both edges of the clock signal; whereas a standard SDRAM performs its operations on the rising edge of the clock signal.

4. *Rambus DRAM (RDRAM)*: The RDRAM provides a very high data transfer rate over a narrow CPU-memory bus. It uses various speedup mechanisms, like synchronous memory interface, caching inside the DRAM chips and very fast signal timing. The Rambus data bus width is 8 or 9 bits.

5. *Cache DRAM (CDRAM)*: This memory is a special type DRAM memory with an on-chip cache memory (SRAM) that acts as a high-speed buffer for the main DRAM.

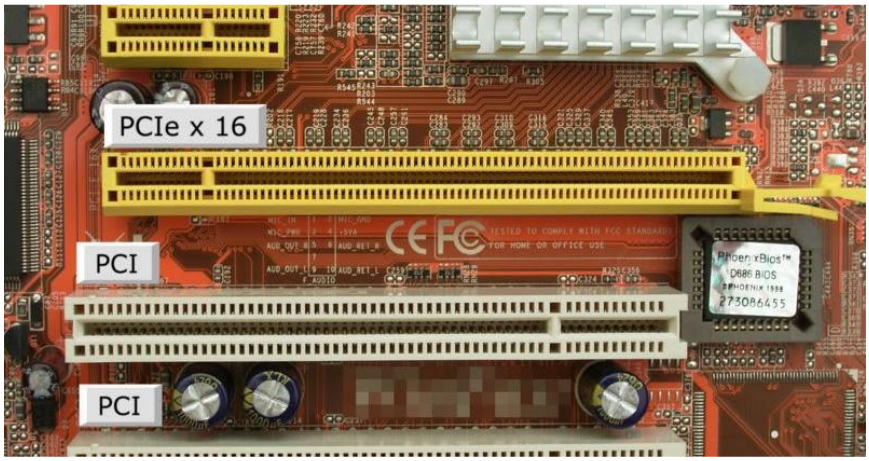
Daughter Cards

A Daughter Card (daughterboard) is an expansion board that connects directly to the motherboard and gives added functionality (e.g., modem). Today, these boards are not found or used in desktop computers. They were replaced with ISA cards, PCI cards, and onboard options. However, some laptops still use these boards.

Bus Slots

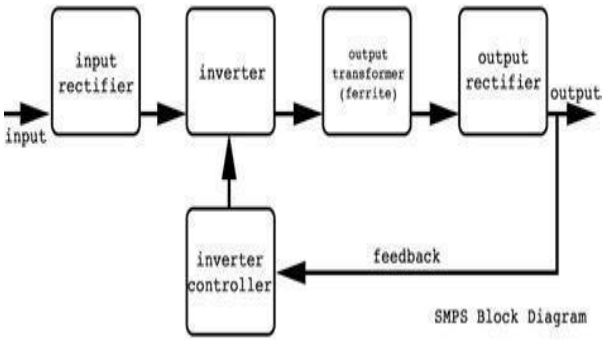
A bus slot or expansion port, an expansion slot is a connection or port inside a computer on the motherboard or riser card. It provides an installation point for a hardware expansion card to be connected. A list of bus slots present commonly in a motherboard are given below:

1. AGP - Video card.
2. AMR - Modem, sound card.
3. CNR - Modem, network card, sound card.
4. ISA - Network card, sound card, video card.
5. PCI - Network card, SCSI, sound card, video card.
6. PCI Express - Video card, modem, sound card, network card.



SMPS

A switched-mode power supply (SMPS) is an electronic circuit that converts power using switching devices that are turned on and off at high frequencies, and storage components such as inductors or capacitors to supply power when the switching device is in its non-conduction state. Switching power supplies have high efficiency and are widely used in a variety of electronic equipment, including computers and other sensitive equipment requiring stable and efficient power supply. A switching regulator does the regulation in the SMPS. A series switching element turns the current supply to a smoothing capacitor on and off. The voltage on the capacitor controls the time the series element is turned. The continuous switching of the capacitor maintains the voltage at the required level.



SMPS Block Diagram

Hard Disk Drive(Magnetic Storage Device)

HDD is an electro-mechanical storage device, which is an abbreviation of Hard Disk Drive. It uses magnetic storage for storing and retrieving the digital data. It is a non-volatile storage device.Hard Disk Drive is installed internally in our computer systems, which is connected directly to the disk controllers of the motherboard. Hard Disk Drive is a storage device which stores the operation system (OS), installed software, and the other computer files.

SSD

A solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies to store data persistently, typically using flash memory, and functioning as secondary storage in the hierarchy of computer storage. It is also sometimes called a solid-state device or a solid-state disk, even though SSDs lack the physical spinning disks and movable read–write heads used in hard disk drives (HDDs) and floppy disks.